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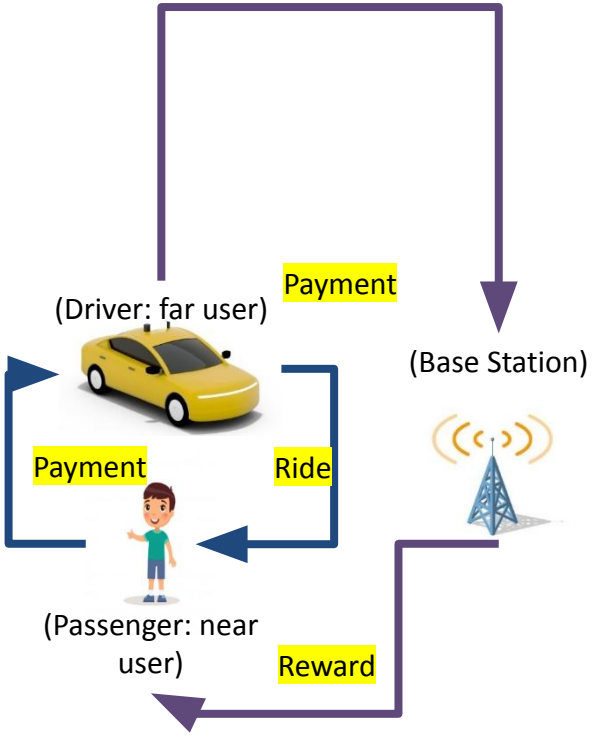


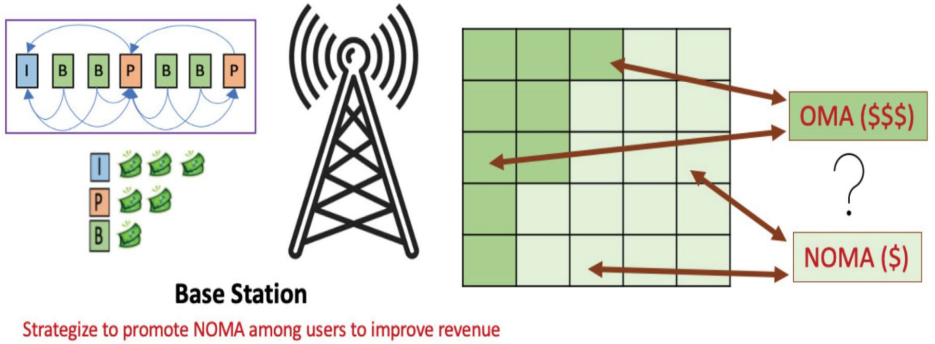


Recent thesis defenses: Jefferson Duan (M.S. 2026) and Yahui Jiang (M.S. 2026).
Left: Vidas (M.S. 2025) Thesis Defense.
Right Top: Krishna Graduation in 2023.
Right Bottom: Evan (B.S. 2022) and Prof. Krishna Ramamoorthy (Ph.D. 2023) in Silicon Valley. Evan is working for META and Krishna is a professor in Santa Clara University.



Uberize Your Wireless Access





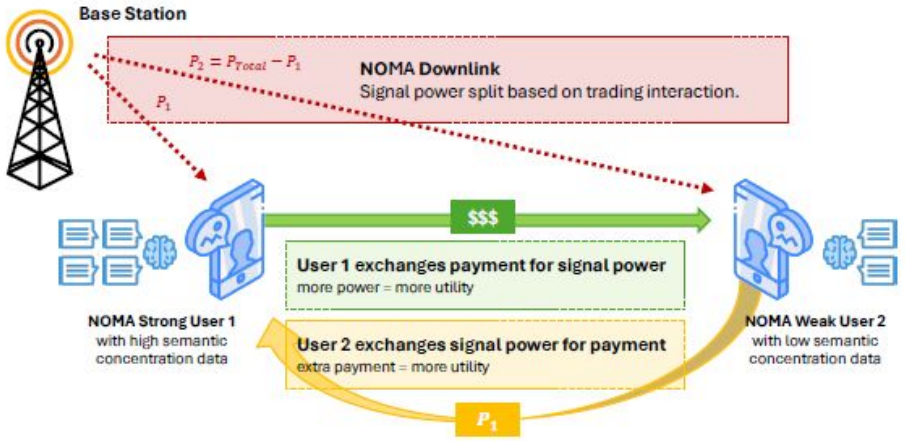


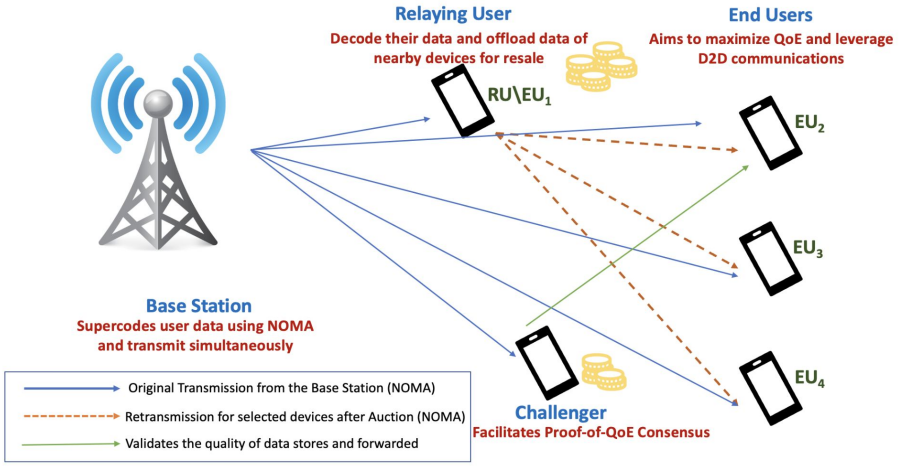
Fig. 1. Semantic-Aware NOMA system model that proposes a power trading relationship between NOMA users

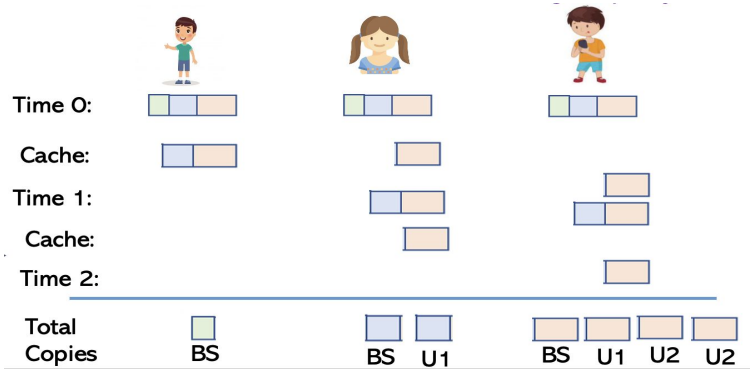
Y. Song (Student Advisee), W. Wang, K. Sohraby, "Uberization of NOMA Wireless Network Resource Sharing: A Driver-Passenger Game-Theoretic Approach," in Proc. IEEE International Workshop on Computer Aided Modeling and Design of Communication Links and Networks (CAMAD), Nov. 2022. **Best Paper Award**.

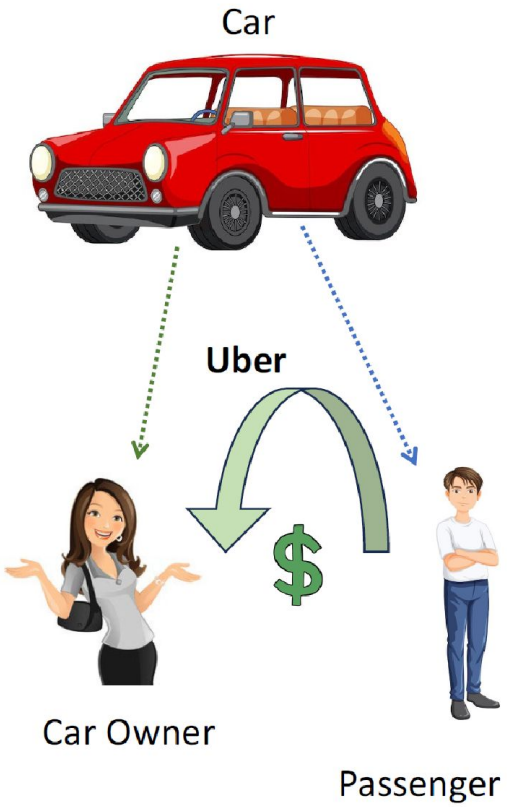
K. Ramamoorthy, W. Wang, K. Sohraby "Incentivize Non-Orthogonal Multiple Access In Wireless Multimedia Communications," in Proc. IEEE Wireless Communications and Networking Conference (WCNC) Workshop on Next Generation Multiple Access for Future Wireless Communications, Mar. 2023.

J. Duan (Student Advisee), K. Ramamoorthy, Y. Zhao, W. Wang, "Stackelberg Game-Theoretic Semantic-Aware Power Trading Models for Safety-Critical NOMA Wireless Systems," to be submitted to IEEE ICC 2027.

Local Wireless Ecosystem: Uberize Token, Pinch Wireless Access







K. Ramamoorthy (Student Advisee), W. Wang, K. Sohraby, Y. Zhao, “Proof-of-QoE NOMA Token: A Crypto Rewarding Concept To Incentivize Local Relay In Non-Orthogonal Multiple Access Wireless Networks,” in Proc. International Conference on Computing, Networking and Communications (ICNC) Workshop on Computing, Networking and Communications (CNC), Feb. 2024.

D. Chui (Student Advisee), W. Wang, K. Ramamoorthy, “Semantic-Aware Learning-Based NOMA User Grouping for Programmable 6G RANs,” in Proc. IEEE Wireless Communications and Networking Conference Workshops (WCNC), Workshop on Intelligent and Programmable Open 6G Radio Access Networks (INTRAN), Apr. 2026.

Intelligent Wireless Networks Research From Bits To Semantics

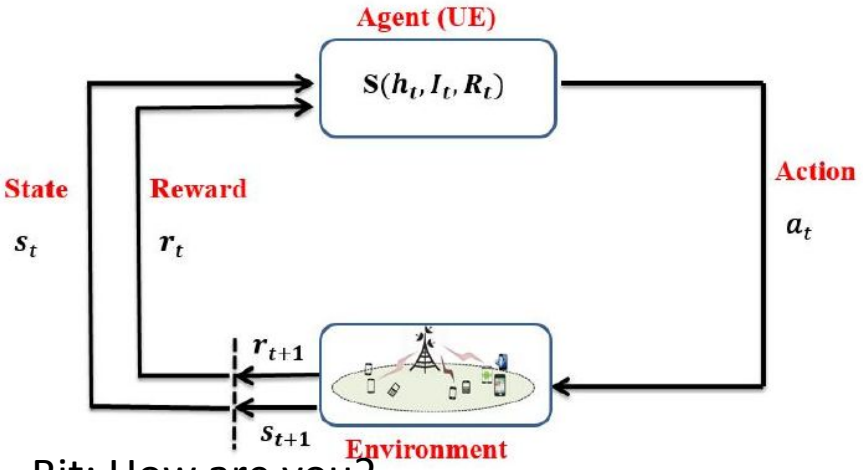




Fig. 2. Screenshot of AOM AV1 analyzer software used to examine details of the frame references.

Bit: How are you?
Semantic: h r u?

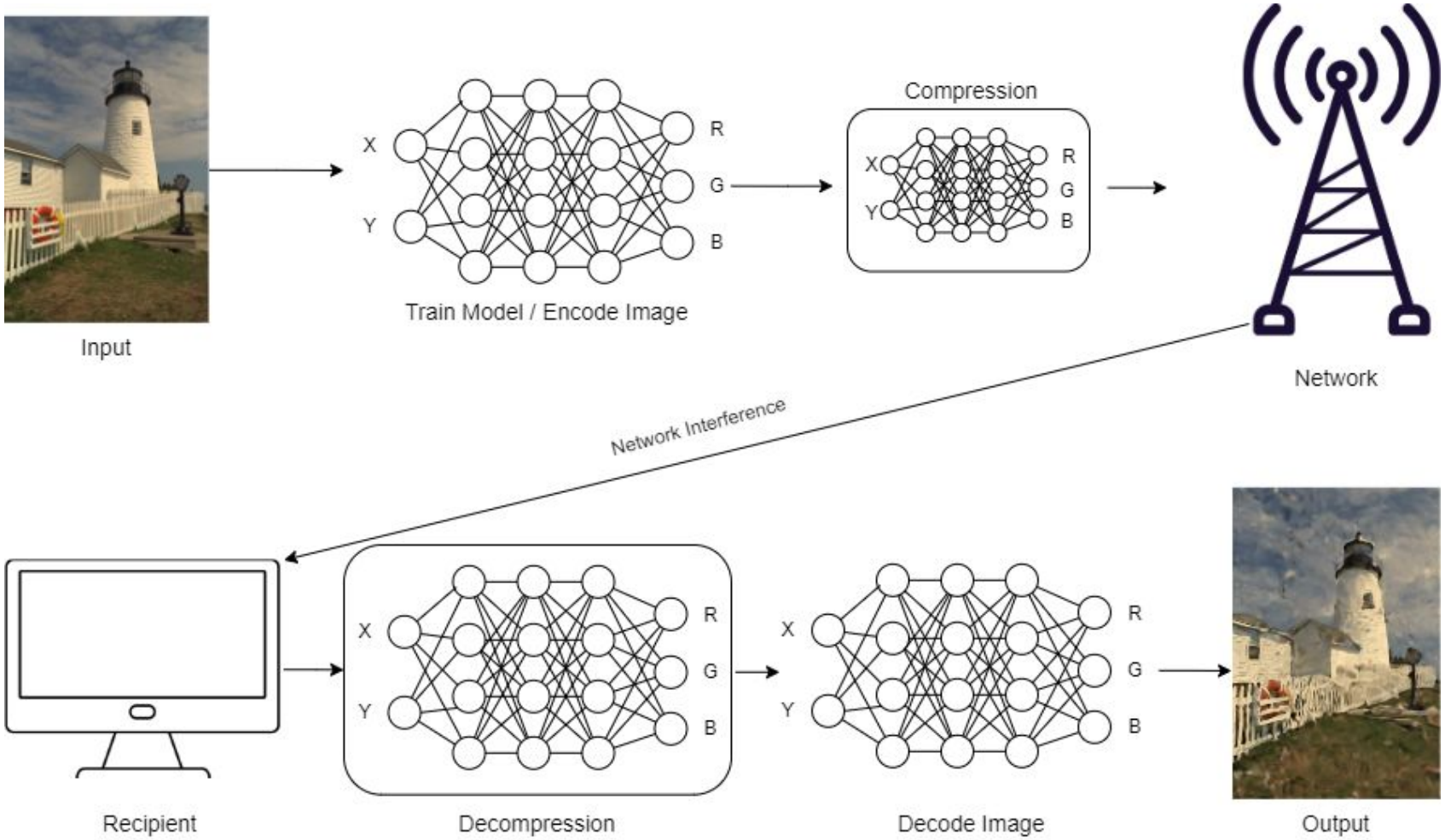
Data is top heavy: 6% data controls 30% quality

On going: Machine learning in wireless semantic communication networks: GAN, RL, **Implicit Neural Representation**, etc.

S. He (Student Advisee), W. Wang, "Pricing QoE With Reinforcement Learning For Intelligent Wireless Multimedia Communications," in Proc. IEEE International Conference on Communications (ICC), Jun 2020.

E. Ballesteros (Student Advisee), K. Ramamoorthy (Student Advisee); W. Wang, "Exploring AV1 Encoder Potentials for Priority-Driven Wireless Multimedia Services," IEEE iETC Conference, May 2022.

Intelligent Wireless Multimedia Networks: Model as Data



T. Trinh, K. Ramamoorthy, I. Koberidze, D. Chui, M. Tsintsadze, Y. Zhao, W. Wang, Erasure-Resilient Image Delivery Using Integer-Quantized Implicit Neural Representations, submitted to IEEE CCNC 2027 under review.